DMA in STM32F103 and Computer Architectures

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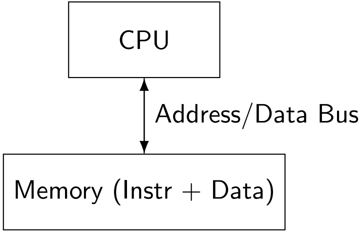
October 3, 2025

Outline

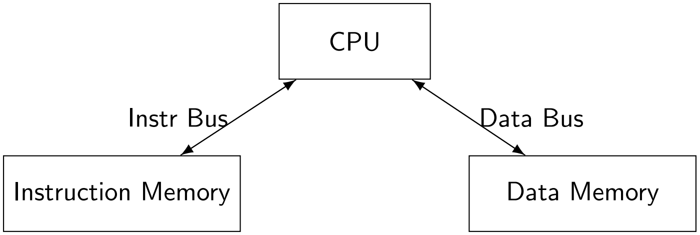
Von Neumann Architecture

- Single memory for instructions and data.
- One shared bus $\rightarrow$ simple, low cost.
- Instructions and data cannot be fetched simultaneously.
- Used in traditional microprocessors.



Harvard Architecture

- Separate memories for instructions and data.
- Separate buses allow simultaneous instruction fetch and data access.
- Higher throughput, more complex.
- Common in microcontrollers and DSP.



Modified Harvard Architecture in STM32

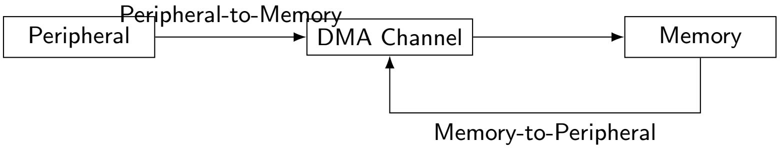
- ARM Cortex-M3 uses a modified Harvard architecture.
- Separate I-Bus, D-Bus, and S-Bus for parallel access.
- Prefetch and pipeline improve instruction throughput.
- Unified memory map seen by the programmer.

Direct Memory Access (DMA) - Overview

- Allows peripherals to access memory without CPU intervention.
- Frees up CPU and increases data throughput.
- Typical use cases:
  - ADC $\rightarrow$ Memory
  - Memory $\rightarrow$ UART
  - SPI $\rightarrow$ Memory

DMA Controller in STM32F103

- DMA1 controller with multiple channels.
- Each channel is assigned to a specific peripheral.
- Supports Peripheral-to-Memory, Memory-to-Peripheral, and Memory-to-Memory.
- Configurable data size, increment mode, circular mode, and priority.



DMA Configuration Steps (HAL)

1. Enable DMA clock.
2. Configure DMA handle: channel, direction, buffer addresses, data length.
3. Link DMA handle to peripheral handle (e.g., ADC).
4. Configure NVIC for DMA interrupts if used.
5. Start DMA transfer using HAL function.

Example: ADC to Memory DMA

```
#define BUFFER_SIZE 100
uint16_t adc_buffer[BUFFER_SIZE];

HAL_ADC_Start_DMA(&hadc1, (uint32_t*)adc_buffer, BUFFER_SIZE);
```

- ADC conversions are automatically stored in adc_buffer.
- CPU can process data while DMA is running.

DMA Interrupts

- Transfer Complete (TC): when DMA finishes all transfers.
- Half Transfer (HT): useful for double-buffering.
- Transfer Error (TE): indicates misconfiguration or bus errors.
- NVIC must be configured to handle these interrupts.

CPU vs DMA Transfers

- CPU polling or interrupt-based transfers consume cycles.
- DMA offloads transfer, enabling parallel CPU tasks.
- Significantly improves throughput for large data blocks.

Practical Applications

- Real-time data acquisition from ADC at high sampling rates.
- Continuous UART transmission of sensor data.
- SPI-based memory logging without CPU bottlenecks.

Summary

- Von Neumann vs Harvard architectures affect memory access strategy.
- STM32 uses modified Harvard architecture with multiple buses.
- DMA is essential for efficient data movement in embedded systems.
- Proper configuration enables high-performance real-time applications.